



ASSIGNMENT II

Association Rule Learning on the Groceries Dataset

Unsupervised Learning Method

Assignment Report

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Software used: Python (pandas, mlxtend, matplotlib)

Question 1: Understand the Data

The Groceries dataset contains real-world point-of-sale transactions from a grocery store, stored in basket format where each row lists the items purchased together in a single shopping trip.

Metric	Value
Number of transactions	9,835
Number of distinct items	169

Top 10 Most Frequently Purchased Items

Rank	Item	Number of transactions
1	whole milk	2,513
2	other vegetables	1,903
3	rolls/buns	1,809
4	soda	1,715
5	yogurt	1,372
6	bottled water	1,087
7	root vegetables	1,072
8	tropical fruit	1,032
9	shopping bags	969
10	sausage	924

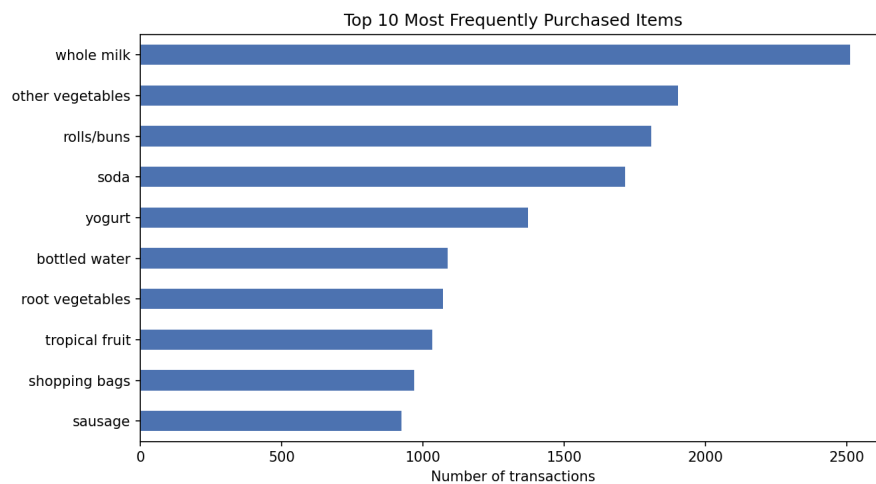


Figure 1: Top 10 most frequently purchased items.

Whole milk is purchased in about 25.6% of all transactions, making it by far the single most common item, followed by other vegetables and rolls/buns. These are staple, high-frequency-purchase items, which is typical of grocery data.

Question 2: Frequent Itemsets

Using the Apriori algorithm with a minimum support threshold of 1% (0.01), 245 frequent itemsets containing two or more items were identified out of 333 total frequent itemsets. The 15 itemsets with the highest support are shown below.

Itemset	Support
other vegetables, whole milk	0.0748
rolls/buns, whole milk	0.0566
whole milk, yogurt	0.0560
root vegetables, whole milk	0.0489
other vegetables, root vegetables	0.0474
other vegetables, yogurt	0.0434
other vegetables, rolls/buns	0.0426
tropical fruit, whole milk	0.0423
soda, whole milk	0.0401
rolls/buns, soda	0.0383
other vegetables, tropical fruit	0.0359
rolls/buns, yogurt	0.0344
bottled water, whole milk	0.0344
pastry, whole milk	0.0332
other vegetables, soda	0.0327

c. Highest-support itemset: {other vegetables, whole milk}, with a support of 0.0748 (about 7.5% of all transactions).

d. What the support value tells us: Support measures how frequently an itemset appears across all transactions. A support of 0.0748 for {other vegetables, whole milk} means that roughly 1 in every 13 shopping trips includes both items together. High-support itemsets like this reveal the store's staple combinations — items that a large share of the customer base buys together regularly, largely because they are everyday household essentials rather than niche or seasonal products.

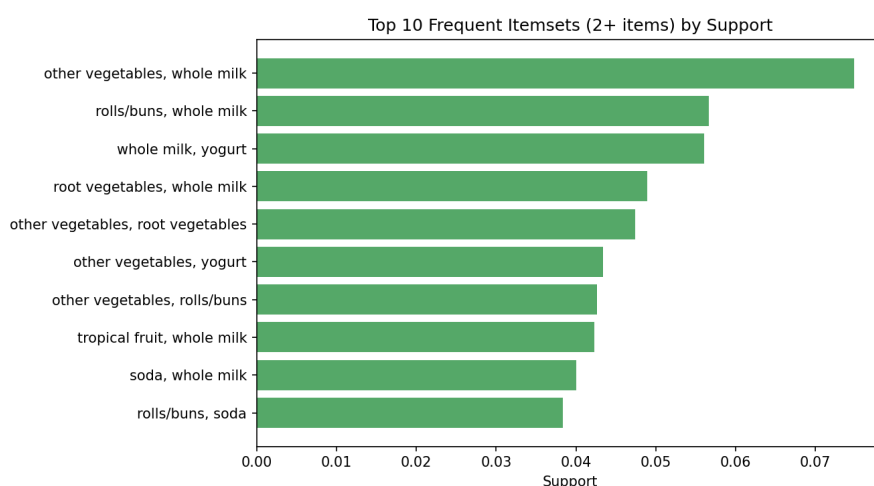


Figure 2: Top 10 frequent itemsets (2+ items) ranked by support.

Question 3: Association Rules

Association rules were generated from the frequent itemsets using a minimum confidence threshold of 0.2, keeping only rules with lift greater than 1 (i.e., rules describing a genuine positive association rather than a coincidental one). This produced 231 rules in total; the top 10 ranked by lift are shown below.

Antecedent	Consequent	Support	Confidence	Lift
citrus fruit, other vegetables	root vegetables	0.0104	0.359	3.295
other vegetables, yogurt	whipped/sour cream	0.0102	0.234	3.267
other vegetables, tropical fruit	root vegetables	0.0123	0.343	3.145
beef	root vegetables	0.0174	0.331	3.040
citrus fruit, root vegetables	other vegetables	0.0104	0.586	3.030
root vegetables, tropical fruit	other vegetables	0.0123	0.585	3.021
root vegetables	other vegetables, whole milk	0.0232	0.213	2.842
other vegetables, whole milk	root vegetables	0.0232	0.310	2.842
butter	other vegetables, whole milk	0.0115	0.207	2.771
curd, whole milk	yogurt	0.0101	0.385	2.761

Question 4: Understanding the Measures

Three representative rules are examined in detail:

Rule 1: {citrus fruit, root vegetables} → {other vegetables}
(support 0.0104, confidence 0.586, lift 3.03)

- **Support (0.0104):** about 1.0% of all 9,835 transactions contained citrus fruit, root vegetables, AND other vegetables together.
- **Confidence (0.586):** of the transactions that included both citrus fruit and root vegetables, 58.6% also included other vegetables.
- **Lift (3.03):** customers who buy citrus fruit and root vegetables together are about 3 times more likely to also buy other vegetables than a random customer — a strong positive association.

Rule 2: {beef} → {root vegetables}
(support 0.0174, confidence 0.331, lift 3.04)

- **Support (0.0174):** about 1.7% of transactions contained both beef and root vegetables.
- **Confidence (0.331):** 33.1% of transactions that included beef also included root vegetables.
- **Lift (3.04):** beef buyers are roughly 3 times more likely than average to also buy root vegetables (e.g., for a stew or roast dinner), indicating a meaningful, non-coincidental relationship.

Rule 3: {other vegetables, whole milk} → {root vegetables}
(support 0.0232, confidence 0.310, lift 2.84)

- **Support (0.0232):** about 2.3% of all transactions contained other vegetables, whole milk, and root vegetables together.
- **Confidence (0.310):** among transactions with both other vegetables and whole milk, 31.0% also contained root vegetables.
- **Lift (2.84):** this combination occurs almost 3 times more often than would be expected if the items were purchased independently, confirming a genuine cross-purchasing pattern among fresh-produce and dairy shoppers.

Question 5: Find Interesting Rules

Criterion	Rule	Support	Confidence	Lift
Highest confidence	{citrus fruit, root vegetables} $\rightarrow$ {other vegetables}	0.0104	0.586	3.030
Highest lift	{citrus fruit, other vegetables} $\rightarrow$ {root vegetables}	0.0104	0.359	3.295
Highest support	{other vegetables} $\rightarrow$ {whole milk}	0.0748	0.387	1.514

d. **Are these the same rule?** No. Each metric highlights a different property, so they identify three different rules:

- The highest-support rule ({other vegetables} $\rightarrow$ {whole milk}) simply reflects that both items are purchased extremely often overall — it is driven by popularity, and its lift (1.51) is comparatively modest, meaning the association is only mildly stronger than chance.
- The highest-confidence and highest-lift rules both involve citrus fruit and root vegetables/other vegetables — smaller, more specific itemsets that are purchased together unusually often relative to how often each item appears alone, even though their overall support is low ($\sim 1\%$).
- This illustrates a key lesson in association rule mining: support, confidence, and lift measure different things (frequency, reliability, and strength-of-association respectively), so the “best” rule depends on which business question is being asked.

Question 6: Interpret a Rule

Selected rule: {citrus fruit, root vegetables} $\rightarrow$ {other vegetables}

In simple language: Customers who purchase citrus fruit and root vegetables also tend to purchase other vegetables. Specifically, 58.6% of shoppers who bought both citrus fruit and root vegetables in the same trip also picked up other vegetables, and this combination is about 3 times more likely than would be expected if the items were chosen independently.

Strength of the association: This is a strong association. A lift of 3.03 (well above the neutral value of 1) shows the items are purchased together far more often than random chance would predict, and a confidence of 58.6% means the rule is right more often than not whenever its antecedent occurs. The main caveat is that its support is low (about 1%), so while the relationship is strong, it applies to a relatively small slice of overall transactions — these are likely health-conscious or produce-focused shoppers rather than the general customer base.

Question 7: Business Application

a. Products to place close together

Root vegetables and other vegetables should be placed near each other (and near citrus fruit). They appear together with high support (0.047) and strong lift (2.8–3.3 across several rules), and are all part of the same “fresh produce” shopping mission, so adjacent placement matches natural customer behavior and can boost basket size.

b. Possible product bundle

A “Fresh Produce Basket” bundle of citrus fruit + root vegetables + other vegetables could be promoted (e.g., a discounted combo or a recipe-card display for a vegetable soup or stew), since these three items co-occur with high confidence and lift.

c. Example use of association rules for recommendations

At checkout (online) or via a loyalty app, when a customer adds root vegetables and other vegetables to their basket, the system could automatically recommend citrus fruit or beef (“customers who bought these items also bought...”), mirroring the rules with lift > 3 discovered above. This is the same logic used by “frequently bought together” recommendation widgets in e-commerce.

Question 8: Understanding Association

a. Does an association rule prove causation?

No. Association rule mining is a correlational, descriptive technique — it identifies items that co-occur in transactions more often than expected, but it says nothing about why. A high-lift rule such as $\{\text{beef}\} \rightarrow \{\text{root vegetables}\}$ does not mean buying beef causes someone to buy root vegetables; both purchases could be driven by a shared underlying cause (e.g., planning a roast dinner), seasonality, store layout, or simple coincidence in a large dataset. Establishing causation would require controlled experiments (e.g., A/B testing store placement), not observational basket data.

b. Why might a rule have high confidence but low lift?

Confidence only measures how often the consequent appears given the antecedent, without accounting for how common the consequent already is on its own. If the consequent is a very popular item (like whole milk, which appears in about 25.6% of all transactions), then almost any antecedent will have reasonably high confidence for predicting it — simply because whole milk shows up everywhere — even if there is no meaningful relationship. In that case confidence is high, but lift (confidence divided by the consequent’s baseline frequency) stays close to 1, revealing that the antecedent adds little real predictive power over just knowing the consequent’s overall popularity.

c. What does a lift value greater than 1 indicate?

A lift greater than 1 indicates a positive association: the antecedent and consequent occur together more often than would be expected if they were statistically independent. The further above 1, the stronger the association (e.g., a lift of 3 means the items co-occur three times more often than chance would predict). A lift of exactly 1 means no association (independence), and

a lift below 1 indicates a negative association, where the presence of the antecedent makes the consequent less likely.

Appendix: Additional Figures

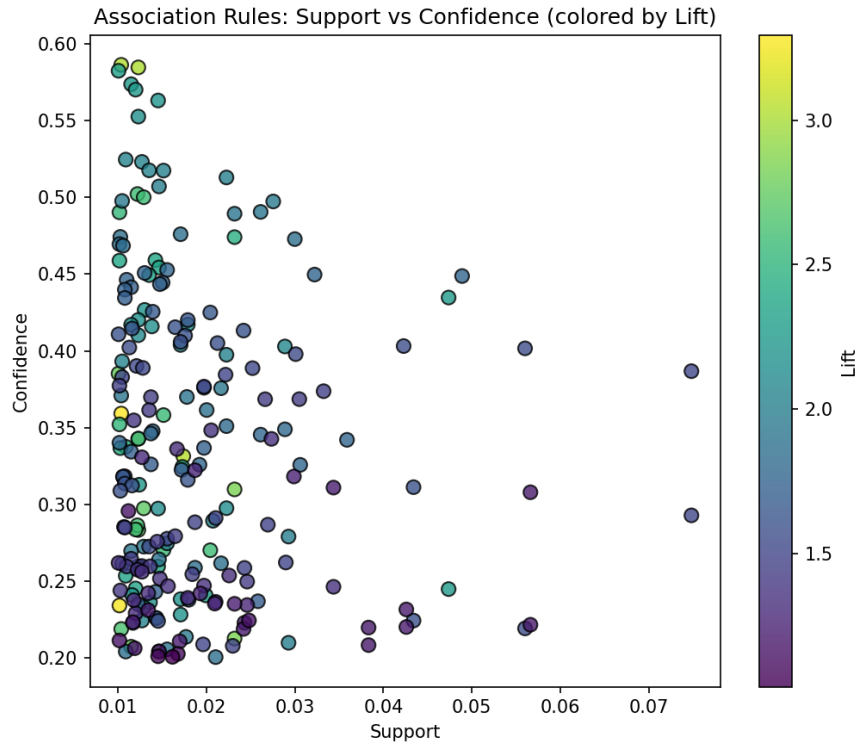


Figure 3: Support vs. confidence for all generated rules, colored by lift.

Methodology Note

Analysis was performed in Python using pandas for data handling, mlxtend (TransactionEncoder, apriori, association_rules) for frequent itemset mining and rule generation, and matplotlib for visualization. Frequent itemsets were mined with a minimum support of 0.01 (1%); association rules were generated with a minimum confidence of 0.2 and filtered to lift > 1 . The full analysis code is provided separately as `analysis.py`.

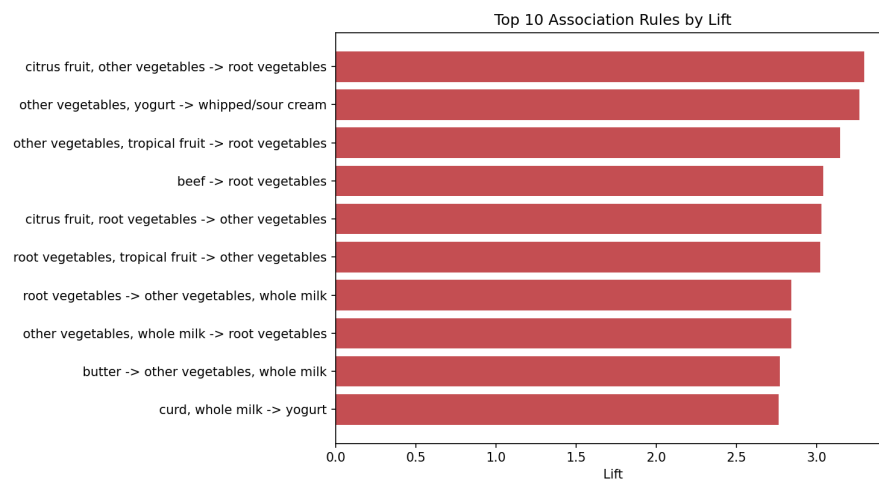


Figure 4: Top 10 association rules ranked by lift.